

ASSIGNMENT #02

Course: CHE 301: CHEMICAL ENGINEERING THERMODYNAMICS
Term: Fall 2025
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Due Date: Wednesday, September 17th in Class (11:00AM)

INSTRUCTIONS FOR ASSIGNMENT DELIVERY

- Prepare a **typed** report of your assignment.
- Include this cover page with your name(s) in the report.
- Upload your report as a PDF file by the Due Date and Time on the UIC Blackboard.
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- Assignments that are not typed will not be accepted.
- Late Assignments will not be accepted without prior approval.
- You may use the Internet as well as ChatGPT or other public tools as long as you place a written statement about their use in your assignment report

CHE 301 HW2

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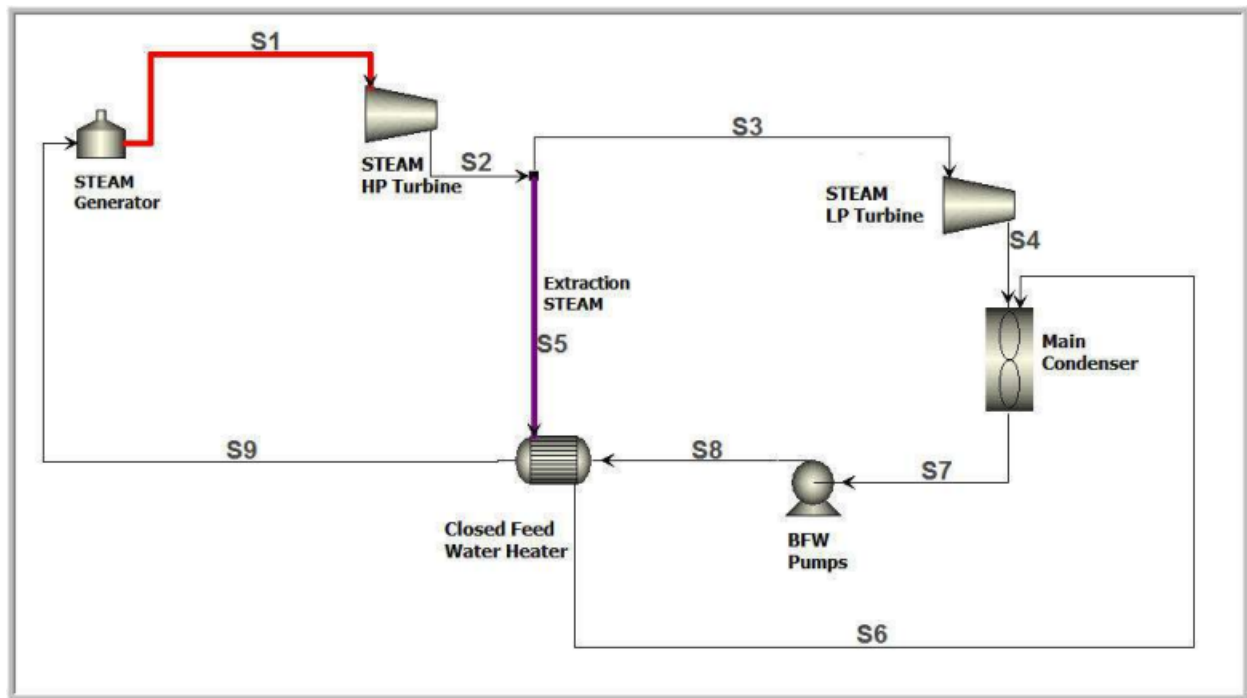


Figure 1: Regenerative Rankine Cycle

1 Equipment Analysis

We believe that performing a complete analysis of all equipments will make it far easier to complete the asked calculations. Thus, the equipment analysis will be in the beginning of our report.

1.1 High Pressure Turbine

The first step in analyzing the high pressure turbine is assuming an initial inlet molar flow rate. We assumed a value of 1 at first. Next we can determine the thermodynamic states of the inlet and outlet streams. It is safe to assume that the inlet stream is superheated

vapor, since it is leaving a steam generator. However, we can confirm this since we know the temperature and pressure of the steam. By using the "Steam Properties" shortcut in the Excel ribbon of the CHE calculator, we can input the given temperature and pressure and see that only values for a vapor state are given, indicating that stream 1, or s_1 is indeed a vapor.

Steam Property	Units	Vapor	Liquid
Pressure	bar	80.000	
Temperature	C	480.000	
Volume	m ³ /kmol	0.726084	
Enthalpy	kJ/kmol	60342.81	
Entropy	kJ/kmol-C	120.0108	
Internal Energy	kJ/kmol	3346.36	

Figure 2: Steam Properties Tool

Next, since the efficiency of the turbine is given, we can utilize a hypothetical, reversible stream in order to solve for the work in a reversible case. This becomes useful, since it allows us to solve for the actual work. Before the actual work can be calculate via the efficiency equation, $s_{2,rev}$ must be analyzed.

First, the thermodynamic state must be designated. In this case, since we are dealing with a reversible case, we know that the entropy generated is zero. Therefore, the entropy of s_1 and $s_{2,rev}$ should be equal. Via the Steam Properties tool, we find that the entropy of a saturated vapor at 15 bar is 116.0272 kJ kmol⁻¹. This is lower than the inlet entropy. In order to increase this value, a greater temperature of this stream is needed, indicating this outlet stream is a superheated vapor.

This "greater temperature" can be calculated via the "goal seek" tool. By setting a cell containing the difference between the entropies of s_1 and $s_{2,rev}$ equal to zero based on the temperature of $s_{2,rev}$, a temperature of 277.8240 °C is obtained.

Vapor and liquid phase enthalpies are calculated via the "HSteam" function. Vapor and liquid phase entropies are calculated via the "Ssteam" function. Although it will not affect

the molar enthalpy or molar entropy, the lever rule was utilized here in order to make future calculations easier. Enthalpy flow and entropy flow are obtained by multiplying the assumed molar flow by their respective molar enthalpies/entropies. Availability is calculated via the "lost work" equation, and power is calculated by taking the difference between the outlet and inlet enthalpies. This gives the following table in the CHE calculator.

Stream Properties	s1	s2,rev
Molar Flow, kmol/s	1.0000	1.0000
Mass Flow, kg/s		
Volume Flow, m ³ /s		
Temperature, C	480.0000	239.7445
Pressure, bar	80.0000	15.0000
Vapor Fraction	1.0000	1.0000
Molecular Weight, kg/kmol		
Molar Enthalpy, kJ/kmol	60342.808	52217.633
Vapor Phase	60342.808	52217.633
Liquid Phase	0.000	0.000
Molar Entropy, kJ/(kmol K)	120.011	120.011
Vapor Phase	120.011	120.011
Liquid Phase	0.000	0.000
Molar Density, kmol/m ³		
Enthalpy Flow, kJ/s	60342.81	52217.63
Entropy Flow, kJ/(s K)	120.01	120.01
Availability, kJ/s	24339.58	16214.33
Duty [Power], kJ/s [kW]		-8125.24

Figure 3: Streams 1 and 2,rev

Now, we must analyze the real stream 2, where the generated entropy is above zero. Considering the molar entropy of the stream can only be greater, it is safe to assume that s_2 is a superheated vapor. Each parameter was calculated in the same fashion for s_1 and $s_{2,rev}$, by using the Ssteam and HSteam functions and lever rule. Now, we can utilize the equation for turbine efficiency.

$$\eta = \frac{W_s}{W_{s,rev}} \quad (1)$$

The given efficiency is 80%. We used goalseek to find a value for the temperature that would force the ratio of W_s and $W_{s,rev}$ to be equal to 0.08. The following table is what results. This concludes the analysis of the high pressure turbine, until later in the report when we have the necessary information for the calculation of the molar flow.

1.2 Low Pressure Turbine

First, it is important to recognize that s_3 splits into two different streams. These streams have different molar flow rates that add up to the initial molar flow rate of s_1 . Thus, we must make a placeholder guess for the molar flow rate that is less than the initial molar flow

Stream Properties	s1	s2,rev	s2
Molar Flow, kmol/s	1.0000	1.0000	1.0000
Mass Flow, kg/s			
Volume Flow, m ³ /s			
Temperature, C	480.0000	239.7445	277.8240
Pressure, bar	80.0000	15.0000	15.0000
Vapor Fraction	1.0000	1.0000	1.0000
Molecular Weight, kg/kmol			
Molar Enthalpy, kJ/kmol	60342.808	52217.633	53842.614
Vapor Phase	60342.808	52217.633	53842.614
Liquid Phase	0.000	0.000	0.000
Molar Entropy, kJ/(kmol K)	120.011	120.011	123.068
Vapor Phase	120.011	120.011	123.068
Liquid Phase	0.000	0.000	0.000
Molar Density, kmol/m ³			
Enthalpy Flow, kJ/s	60342.81	52217.63	53842.61
Entropy Flow, kJ/(s K)	120.01	120.01	123.07
Availability, kJ/s	24339.58	16214.33	16922.08
Duty [Power], kJ/s [kW]		-8125.24	-6500.19
Efficiency			0.80

Figure 4: Streams 1, 2.rev, 2

rate. We chose 0.5 here. The temperature, pressure, and thermodynamic state of s_3 should all be equal to that of s_2 .

We are again given the efficiency of this equipment, meaning we can take a very similar approach to the high pressure turbine analysis. Again, imagine a reversible process with a hypothetical outlet stream. Then, analyze the thermodynamic states. Using the Steam Properties tool, we can see that the entropy of saturated vapor at 0.08 bar is greater than that of the inlet stream. Unfortunately, this indicates that the stream is a mixture, since a more ordered thermodynamic state, such as liquid, must be present in the stream in order for the entropy to be a lower value.

As a result, we do not need to make a guess for the temperature, but rather the vapor fraction, or quality of the mixture. We make the guess 0.5 here. Molar enthalpy and molar entropy are calculated using the lever rule. Vapor phase and liquid phase quantities were obtained via the same aforementioned CHE functions. Each parameter is calculated the same, then goalseek is utilized to find a value of the quality that forces the inlet and outlet entropies to be equal. Here, this value turns out to be 0.8169 for $s_{4,rev}$. Then, the same process used to determine the temperature of s_2 is utilized to determine the quality of s_4 . The following table is given

Stream Properties	s3	s4,rev	s4
Molar Flow, kmol/s	0.5000	0.5000	0.5000
Mass Flow, kg/s			
Volume Flow, m ³ /s			
Temperature, C	277.8240		
Pressure, bar	15.0000	0.0800	0.0800
Vapor Fraction	1.0000	0.8169	0.8878
Molecular Weight, kg/kmol			
Molar Enthalpy, kJ/kmol	53842.614	38500.526	41568.944
Vapor Phase	53842.614	46426.672	46426.672
Liquid Phase	0.000	3132.177	3132.177
Molar Entropy, kJ/(kmol K)	123.068	123.068	132.819
Vapor Phase	123.068	148.256	148.256
Liquid Phase	0.000	10.675	10.675
Molar Density, kmol/m ³			
Enthalpy Flow, kJ/s	26921.31	19250.26	20784.47
Entropy Flow, kJ/(s K)	61.53	61.53	66.41
Availability, kJ/s	8461.04	790.06	861.65
Duty [Power], kJ/s [kW]		-7671.04	-6136.84
Efficiency			0.80

Figure 5: LP Turbine Streams

1.3 Boiling Feed Water Pump

For any pump, the input and output are both entirely liquid. Additionally, the inlet stream temperature is given, and the efficiency of the pump is given. The pressure is also known, since pressure is constant through the main condenser. Once again, the same process of using an imaginary, reversible stream can be utilized. This is because we know the efficiency. The process is the same as the turbines, except that the equation for the efficiency is flipped, with the reversible work being divide by the actual work. As a result, this table is obtained.

Stream Properties	s7	s8,rev	s8
Molar Flow, kmol/s	1.0000	1.0000	1.0000
Mass Flow, kg/s			
Volume Flow, m ³ /s			
Temperature, C	38.0000	38.2229	39.0500
Pressure, bar	0.0800	80.0000	80.0000
Vapor Fraction	0.0000	0.0000	0.0000
Molecular Weight, kg/kmol			
Molar Enthalpy, kJ/kmol	2866.106	3010.705	3072.676
Vapor Phase	0.000	0.000	0.000
Liquid Phase	2866.106	3010.705	3072.676
Molar Entropy, kJ/(kmol K)	9.824	9.824	10.023
Vapor Phase	0.000	0.000	0.000
Liquid Phase	9.824	9.824	10.023
Molar Density, kmol/m ³			
Enthalpy Flow, kJ/s	2866.11	3010.70	3072.68
Entropy Flow, kJ/(s K)	9.82	9.82	10.02
Availability, kJ/s	-81.23	63.50	65.85
Duty [Power], kJ/s [kW]		144.60	206.57
Efficiency			0.70

Figure 6: BFW Streams

1.4 Closed Feed Water Heater

Assume that there is no heat transfer between the streams and the environment. The closed feed water heater is essentially a shell-tube heat exchanger, or a heat exchanger without mixing. There are two streams coming in, and two streams going out. Each inlet stream has a corresponding outlet stream of equal absolute value. Inlet and outlet enthalpies are all available based on information given in the problem. H_5 is equal to H_2 . It is given in the problem that s_6 is a subcooled liquid. Assuming constant pressure across the closed feed water heater, H_6 can be calculated via the Steam Properties tool. It is also given that the temperature of s_9 is 5 degrees Celsius than that of s_6 . Assuming s_9 is a subcooled liquid and constant pressure across the closed feed water heater, the Steam Properties tool can be used to get H_9 . These values are expressed in the table below.

Stream Properties	s5	s6	s8	s9
Molar Flow, kmol/s	0.5000	0.5000	1.0000	1.0000
Mass Flow, kg/s				
Volume Flow, m ³ /s				
Temperature, C	277.8240	198.2890	39.0500	193.2890
Pressure, bar	15.0000	15.0000	80.0000	80.0000
Vapor Fraction	1.0000	0.0000	0.0000	0.0000
Molecular Weight, kg/kmol				
Molar Enthalpy, kJ/kmol	53842.614	15216.608	3072.676	14866.514
Vapor Phase	53842.614	0.000	0.000	0.000
Liquid Phase	0.000	15216.608	3072.676	14866.514
Molar Entropy, kJ/(kmol K)	123.068	41.695	10.023	40.662
Vapor Phase	123.068	0.000	0.000	0.000
Liquid Phase	0.000	41.695	10.023	40.662
Molar Density, kmol/m ³				

Figure 7: CFW Streams

1.5 Full Material and Energy Balance

Now, we know enough information in order to solve for the molar flow of the system. We created two cells, one representing the heat exchange of the feed water, and another representing the heat exchange of the extraction steam. We decided to vary s_1 and s_3 so that the total power of both turbines combines to 1000 MW, and so that the two cells representing the heat exchange in the closed feed water heater are equal to each other. This gives the following table.

Stream Properties	s1	s2,rev	s2	s3	s4,rev	s4	s5	s6	s7	s8,rev	s8	s9
Molar Flow, kmol/s	62.9148	62.9148	62.9148	48.1552	48.1552	48.1552	14.7596	14.7596	62.9148	62.9148	62.9148	62.9148
Mass Flow, kg/s												
Volume Flow, m ³ /s												
Temperature, C	480.0000	239.7445	277.8240	277.8240	41.5340	41.5340	277.8240	198.2890	38.0000	38.2229	39.0500	193.2890
Pressure, bar	80.0000	15.0000	15.0000	15.0000	0.0800	0.0800	15.0000	15.0000	0.0800	80.0000	80.0000	80.0000
Vapor Fraction	1.0000	1.0000	1.0000	1.0000	0.8169	0.8878	1.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Molecular Weight, kg/kmol												
Molar Enthalpy, kJ/kmol	60342.808	52217.633	53842.614	53842.614	38500.526	41568.944	53842.614	15216.608	2866.106	3010.705	3072.676	14866.514
Vapor Phase	60342.808	52217.633	53842.614	53842.614	46426.672	46426.672	53842.614	0.000	0.000	0.000	0.000	0.000
Liquid Phase	0.000	0.000	0.000	0.000	3132.177	3132.177	0.000	15216.608	2866.106	3010.705	3072.676	14866.514
Molar Entropy, kJ/(kmol K)	120.011	120.011	123.068	123.068	123.068	132.819	123.068	41.695	9.824	9.824	10.023	40.662
Vapor Phase	120.011	120.011	123.068	123.068	148.256	148.256	123.068	0.000	0.000	0.000	0.000	0.000
Liquid Phase	0.000	0.000	0.000	0.000	10.675	10.675	0.000	41.695	9.824	9.824	10.023	40.662
Molar Density, kmol/m ³												
Enthalpy Flow, kJ/s	3796457.69	3285263.68	3387499.08	2592803.39	1854001.66	2001762.00	3387499.08	224591.12	180320.60	189418.00	193316.89	935324.26
Entropy Flow, kJ/(s K)	7550.46	7550.47	7742.83	5926.39	5926.37	6395.92	7742.83	615.41	618.10	618.08	630.58	2558.27
Availability, kJ/s	1531320.39	1020122.09	1064649.60	814886.45	76091.35	82986.22	1064649.60		-5110.63	3995.40	4142.79	
Duty [Power], kJ/s [kW]		-511198.30	-408958.61		-738801.74	-591041.39				9097.40	12996.29	
Efficiency			0.80			0.80					0.70	

Figure 8: Full Balance

2 Calculations

2.1 The amount of steam required in MMSCFD

The overall molar flow rate, or the flow rate of s_1 simply needs to be converted into MMSCFD. In the CHE calculator, the function "KMOLperSECtoMMSCFD" makes this conversion trivial. We obtained the following value: 4547.6571 MMSCFD

2.2 The amount of Bituminous coal required in kg/s

To solve this, some analysis of the boiler is necessary. An energy balance of the boiler gives the equation:

$$\dot{Q} = \dot{m}(h_{out} - h_{in}) \quad (2)$$

Substituting values from the overall mass and energy balance gives

$$\dot{Q} = 62.9148(60342.808 - 14866.514) = 2861133.4265 \text{kJ/s} \quad (3)$$

Now, we must calculate the energy required based on $\dot{Q}$ and η . We can use this equation:

$$\dot{E}_{\text{fuel}} = \frac{\dot{Q}_{\text{boiler}}}{\eta_{\text{boiler}}}$$

Which evaluates as

$$\frac{2861133.4265}{0.8} = 3576416.7831 \text{kJ/s} \quad (4)$$

Now, divide by the heating value of coal

$$\frac{3576416.7831}{22000} = 162.56 \text{kg/s} \quad (5)$$

Our final answer is 162.56 kg/s

2.3 The lost work in each equipment of the Regenerative Rankine Cycle

From the text book, the steady state entropy balance is

$$-m\Delta S + \frac{\dot{Q}_1}{T_1} + \frac{\dot{Q}_2}{T_2} + \frac{\dot{Q}_0}{T_0} + \dot{S}_{\text{gen}} = 0$$

Assuming the high pressure turbine is insulated, the equation can simplify to

$$\dot{S}_{\text{gen}} = m\Delta S \quad (6)$$

Which can be substituted into the lost work equation to give

$$T_0 \dot{S}_{\text{gen}} = T_0 \dot{m}(s_{out} - s_{in})$$

where s represents the entropy, rather than referencing the stream as a whole.

Evaluating this for the high pressure turbine gives:

$$(300)(62.9148)(123.068 - 120.011) = 57,699.16\text{kJ/s} \quad (7)$$

Following the same logic for the low pressure turbine gives

$$(300)(48.1552)(132.819 - 123.068) = 140,868.41\text{kJ/s} \quad (8)$$

For the condenser, the lost work is equal represented by the equation below:

$$\dot{m}_1[(H_4 + H_6) - T_o(s_4 + s_6) - (H_7 - T_o s_7)] \quad (9)$$

This evaluates to

$$62.9148[(15216.608 + 41568.944) - 300(132.819 + 41.695) - (2866.106 - 300(9.824))] = 283,899.64\text{kJ/s} \quad (10)$$

For a pump, the lost energy is evaluated in the same way as a turbine. Using the earlier equation, this evaluates

$$(300)(62.9148)(10.023 - 9.824) = 3756.01\text{kJ/s} \quad (11)$$

For the closed feed water heater, entropy generation for the feed water and for the steam is calculated separately then summed up. The equation looks like this

$$\dot{S}_{gen} = \dot{m}_{fw}(s_{fw,out} - s_{fw,in}) + \dot{m}_{steam}(s_{steam,out} - s_{steam,in})$$

Which would then be multiplied by the ambient temperature to get the lost work. It evaluates like this

$$(300)[14.7596(41.695 - 123.068) + 48.1552(40.662 - 10.023)] = 82,318.27\text{kJ/s} \quad (12)$$

For the steam generator, the heat input must be factored into the entropy balance. Working this into the lost work equation yields this

$$T_0 \left[\dot{m}(s_{out} - s_{in}) - \frac{\dot{Q}_{in}}{T_b} \right]$$

This evaluates to

$$300[62.9148(120.011 - 40.662) - \frac{2861133.4265}{1300}] = 837,406.38\text{kJ/s} \quad (13)$$

To summarize, the lost work of each equipment is equal to:

HP Turbine = 57,699.16kJ/s

LP Turbine = 140,868.41kJ/s

Condenser = 83,899.64kJ/s

Pump = 3756.01kJ/s

Closed Feed Water Heater = 82,318.27kJ/s

Steam Generator = 837,406.38kJ/s

2.4 The thermal and thermodynamic efficiency of the cycle if the goal is power generation

The thermal efficiency of the cycle is represented by this equation

$$\eta = \frac{\dot{W}_{net}}{\dot{Q}_{in}}$$

$\dot{W}_{net}$ is represented by the total power produced by the turbines subtracted by the total power put into the pump. In this plant, that evaluates to this

$$1000000 - 12996.29 = 987003.71\text{kW}$$

$\dot{Q}_{in}$ is represented by the product of the molar flow rate and the difference of outlet and inlet enthalpies of the boiler. This was evaluated earlier. So, the thermal efficiency can be calculated

$$\eta = \frac{987003.71}{2861133.4265} * 100\% = 34.50\%$$

The thermodynamic efficiency can be represented as the ratio of thermal efficiency to the Carnot efficiency. The Carnot efficiency is represented by the equation

$$1 - \frac{T_o}{T_H}$$

In this case, the Carnot efficiency can be evaluated as

$$1 - \frac{300}{1300} = 0.769$$

Which allows the thermodynamic efficiency to be calculated as

$$\frac{0.3450}{0.7690} = 0.449$$

The thermal efficiency is 34.5%, and the thermodynamic efficiency is 44.9%

2.5 The emissions of CO2 in gigatons per year

Asking ChatGPT, burning one kilogram of bituminous coal creates around 2.5 kilograms of CO2. Assuming a coal plant is operating 90% of the year, stopping occasionally for maintenance, it is running for 7884 hours a year. Using the amount of bituminous coal required for 1000 MW of power, we utilized dimensional analysis to provide an estimate of CO2 emissions in gigatons per year.

$$162.56\text{kg/s} * \frac{3600\text{s}}{1\text{hr}} * \frac{7884\text{hr}}{1\text{year}} * \frac{12 * 10^{-12}\text{gigatons}}{1\text{kg}} = 0.055\text{gigaton/year}$$

CO2 emissions is 0.055 gigatons per year

2.6 Find through research the CO2 emissions per year worldwide in gigatons

In 2024, the global CO2 emissions was 37.41. This information was found on <https://www.statista.com/statistics/co2-emissions/>